Supplementary material

Does starting age predict who stays? Competition debut and long-term participation in youth track and field

Anonymised manuscript. Tables S1–S4 are numbered in the order of first citation in the main text; Table S5 collects the robustness analyses cited in Sections 3.2 and 3.5.

Table S1. Sample flow.

Step	n
Athletes born 2001–2008 in the registry extract	33 666
Result rows extracted (2011 – 29 Oct 2026)	675 707
excluded: non-approved rows	1 526
excluded: rows below age 10	116
Debutants with an approved result at age $\geq$10	28 339
excluded: missing sex	865
set aside: debut at age $\geq$17 (sensitivity analyses)	731
Debutants at ages 10–16 with known sex	26 743
primary sample: born 2001–2005	17 496
descriptive only: born 2006–2008	9 247
Active seniors in the primary sample	546

Table S2. Senior conversion by debut age and sex (born 2001–2005), % (Wilson 95 % CI).

Debut age	Girls, n	Girls, % (CI)	Boys, n	Boys, % (CI)
10	3 485	3.4 (2.8–4.0)	3 141	4.3 (3.6–5.1)
11	1 860	2.2 (1.6–2.9)	1 756	2.7 (2.1–3.6)
12	1 263	1.8 (1.2–2.7)	1 266	2.3 (1.6–3.3)
13	879	2.0 (1.3–3.2)	812	3.8 (2.7–5.4)
14	789	2.0 (1.3–3.3)	958	2.7 (1.9–3.9)
15	326	2.5 (1.2–4.8)	537	4.8 (3.3–7.0)
16	158	5.7 (3.0–10.5)	266	7.1 (4.6–10.9)

Sex $\times$ debut-age interaction (linear): OR 0.92 per year for girls relative to boys, 95 % CI 0.82–1.02. Full coefficient tables for the primary models are available with the analysis code. Abbreviations: CI, confidence interval; OR, odds ratio.

Table S3. Retention k years after the debut season: proportion (Wilson 95 % CI). Denominators equal the debut-age n except debut 16 at k = 5 (n = 389).

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	.512 (.500–.524)	.379 (.367–.390)	.288 (.277–.299)	.218 (.208–.228)	.169 (.160–.178)
11	3 616	.398 (.382–.414)	.269 (.254–.283)	.188 (.175–.201)	.131 (.121–.143)	.097 (.087–.107)
12	2 529	.372 (.353–.391)	.234 (.218–.251)	.143 (.130–.157)	.094 (.083–.106)	.071 (.061–.081)
13	1 691	.378 (.356–.402)	.226 (.207–.247)	.150 (.134–.168)	.100 (.087–.115)	.068 (.057–.081)
14	1 747	.340 (.318–.363)	.128 (.113–.144)	.080 (.068–.093)	.055 (.045–.067)	.033 (.025–.042)
15	863	.256 (.228–.286)	.125 (.105–.149)	.080 (.064–.100)	.049 (.036–.065)	.044 (.032–.060)
16	424	.349 (.305–.396)	.210 (.174–.251)	.111 (.084–.144)	.066 (.046–.094)	.059 (.040–.087)

Abbreviation: CI, confidence interval.

Table S4. Competing at equal career length: proportion with at least two competition days in a single season at least 4, 5 and 6 years after the debut season, by debut age (n = athletes for whom that season was fully observable by 2025), with the adjusted OR per year of later debut (sex, birth cohort, club size).

Debut age	4 years: % (n)	5 years: % (n)	6 years: % (n)
10	20.4 (6 626)	15.7 (6 626)	12.0 (6 626)
11	12.4 (3 616)	8.8 (3 616)	6.4 (3 616)
12	8.5 (2 529)	6.2 (2 529)	4.0 (2 529)
13	8.7 (1 691)	6.2 (1 691)	4.1 (1 691)
14	4.5 (1 747)	2.9 (1 747)	2.2 (1 747)
15	4.9 (863)	3.8 (863)	2.6 (779)
16	6.6 (424)	5.1 (389)	3.2 (317)
Linear trend, OR per +1 year	0.70 (0.68–0.72)	0.69 (0.66–0.72)	0.66 (0.63–0.69)

Abbreviation: OR, odds ratio.

Table S5. Robustness analyses. Estimates are for debut age (per +1 year, linear) unless stated; outcome is senior activity unless stated. Abbreviations: CI, confidence interval; COVID-19, coronavirus disease 2019; OR, odds ratio; SE, standard error.

Analysis	n	Events	OR (95 % CI)
Primary linear trend	17 496	546	0.99 (0.94–1.04)
Excluding debuts in 2020–2021	17 305	530	0.95 (0.90–1.01)
Debut at 16 vs. debut at 10 (categorical), primary	17 496	546	1.73 (1.15–2.59)
excluding debuts in 2020–2021	17 305	530	1.32 (0.79–2.19)
adding the COVID-19 debut flag	17 496	546	1.38 (0.86–2.19)
cohorts born 2001–2003	10 667	349	1.35 (0.80–2.26)
cohorts born 2004–2005	6 829	197	3.08 (1.59–5.96)
Excluding the 2001 cohort (debut at 10 in 2011)	14 077	434	0.99 (0.94–1.05)
Including athletes without registered sex	18 045	547	0.99 (0.94–1.04)
Outcome: $\geq$1 result at age $\geq$20	17 496	694	1.00 (0.96–1.05)
Outcome: any result at age $\geq$17 (debuts 10–14)	16 209	1 621	0.97 (0.93–1.00)
Adding debut-season competition days	17 496	546	0.94 (0.89–1.00)
Cohorts born 2001–2003	10 667	349	0.98 (0.92–1.05)
Cohorts born 2004–2005	6 829	197	0.99 (0.91–1.08)
Debut at 17–19 vs. debut at 10 (categorical)	17 898	595	3.36 (2.42–4.67)
Adding the COVID-19 debut flag (linear trend)	17 496	546	0.95 (0.90–1.01)
Club-clustered robust SE (trend)	17 496	546	0.99 (0.92–1.06)